Rapido Ride Booking Dashboard Report

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Tools Used: Excel, Power BI

- ## Project Objective

The objective of this project is to analyze ride-booking data to gain meaningful insights into booking patterns, revenue generation, cancellation behaviour, and overall service performance. Using tools like Excel, SQL, and Power BI, the goal is to transform raw data into an interactive dashboard that helps identify trends, measure key performance indicators, and support data-driven decision-making.

This analysis aims to improve operational efficiency, reduce cancellation rates, enhance customer satisfaction, and optimize resource utilization across different vehicle types.

- ## Dataset Description

The dataset used in this project contains detailed ride-booking records for a one-month period (February 2026). It captures end-to-end information about each ride, including booking details, vehicle type, trip metrics, revenue, cancellation reasons, and user ratings. This dataset is designed to provide a comprehensive view of operational performance and customer behaviour.

Each record in the dataset represents a single ride booking and includes key attributes such as booking date, customer ID, vehicle category (Prime Sedan, SUV, Mini, Auto, Bike, E-Bike), booking status (Success, Cancelled, Incomplete, System Cancelled), fare amount, and distance travelled.

Additionally, the dataset includes important fields related to cancellations, such as reasons for cancellation from both the customer and driver perspectives. It also contains rating metrics for both customers and drivers, enabling analysis of service quality and satisfaction levels.

The dataset is well-suited for performing:

- Trend analysis (daily bookings & revenue)
- Performance analysis (success vs cancellation rates)
- Revenue analysis across booking statuses
- Vehicle-wise comparison
- Customer and driver behaviour analysis

Overall, this dataset enables a 360-degree analysis of ride-booking operations and supports the creation of an interactive dashboard for actionable insights.

- ## Data Cleaning & Preparation

- **Data Import & Initial Review**
  Imported the raw dataset into Excel and reviewed the structure, column names, and data types to understand the data quality and identify inconsistencies.

- **Handling Missing Values**
  Identified null or blank values in key columns (such as booking status, fare, and ratings) and handled them using appropriate methods like removal, replacement, or imputation.

- **Removing Duplicates**
  Checked for duplicate records based on booking ID and removed them to ensure data accuracy and avoid double counting.

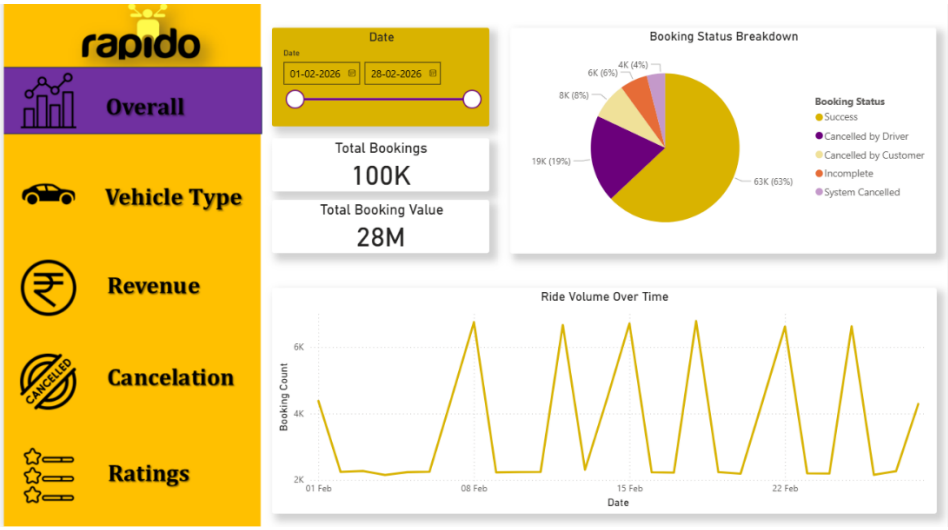
- **Data Formatting & Standardization**
  Standardized column formats such as date, time, and numerical values. Corrected inconsistent naming in categorical fields (e.g., vehicle types, booking status).

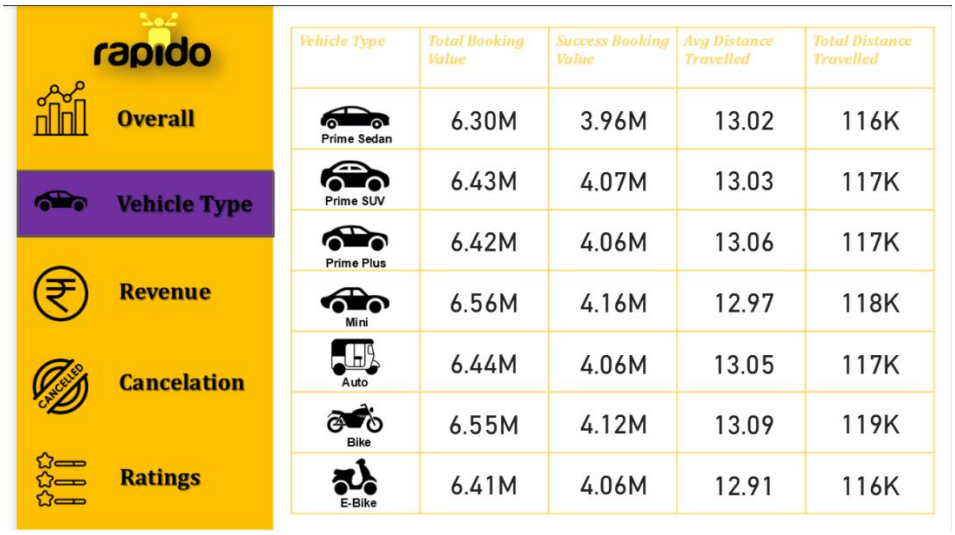
- **Using Power Query for Transformation**
  Utilized Power Query to clean and transform data efficiently by applying steps like filtering, splitting columns, merging data, and creating structured tables.

- **Creating New Columns**
  Derived new fields such as total revenue, cancellation rate, and ride success flag to support better analysis and dashboard creation.

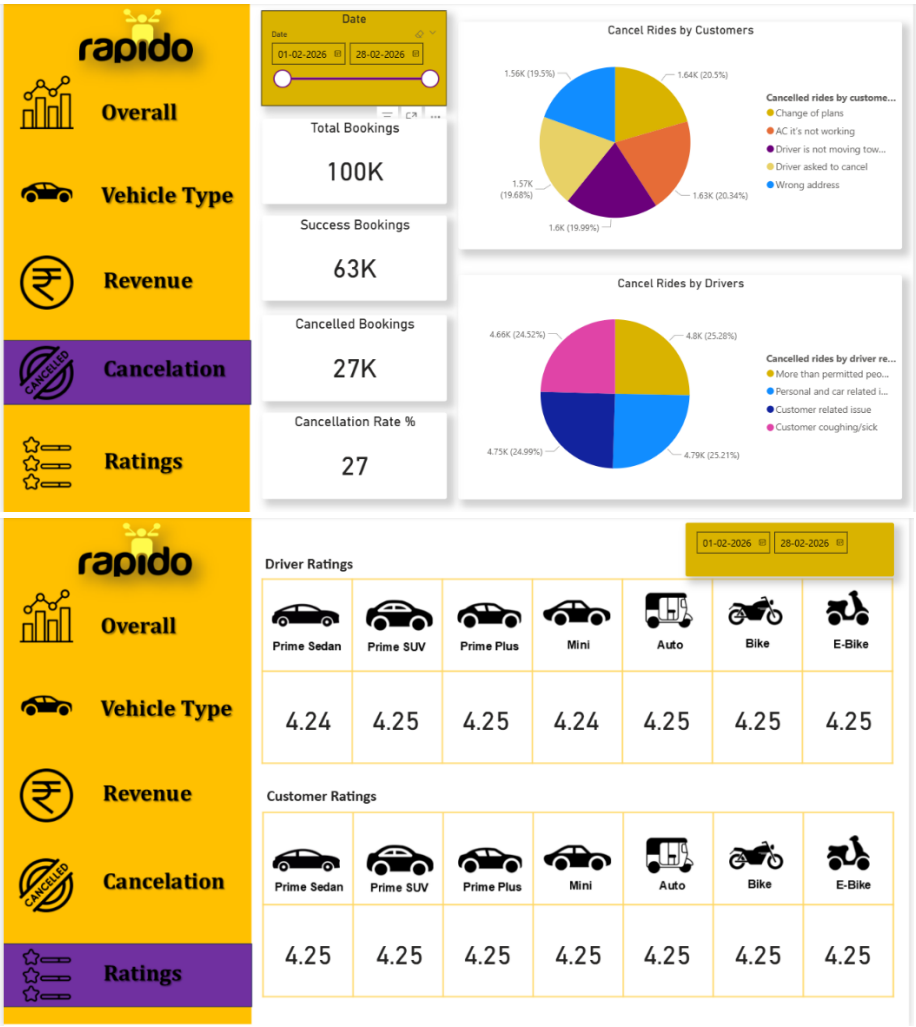
- **Data Validation & Final Checks**
  Performed final validation to ensure data consistency, accuracy, and readiness for visualization in Power BI.

- **Dashboard Overview**





Vehicle Type	Total Booking Value	Success Booking Value	Avg Distance Travelled	Total Distance Travelled
Prime Sedan	6.30M	3.96M	13.02	116K
Prime SUV	6.43M	4.07M	13.03	117K
Prime Plus	6.42M	4.06M	13.06	117K
Mini	6.56M	4.16M	12.97	118K
Auto	6.44M	4.06M	13.05	117K
Bike	6.55M	4.12M	13.09	119K
E-Bike	6.41M	4.06M	12.91	116K



The dashboard is designed to provide a comprehensive and interactive view of ride-booking operations, focusing on key performance indicators such as total bookings, revenue, cancellations, and customer satisfaction. Built using Power BI, the dashboard enables users to quickly understand trends and make data-driven decisions.

At the top level, the dashboard highlights important KPIs including **Total Bookings (100K), Total Revenue (28M), Successful Bookings (63K), and Cancellation Rate (27%)**, giving a quick snapshot of overall performance.

The dashboard is divided into multiple sections for better analysis:

- **Booking Trends Analysis**
  Displays daily booking patterns to identify peak demand periods and fluctuations over time.

- **Booking Status Breakdown**
  Shows the distribution of successful, cancelled, and incomplete rides, helping evaluate operational efficiency.
- **Revenue Analysis**
  Provides insights into revenue contribution by different booking statuses and vehicle types.
- **Vehicle Type Performance**
  Compares ride volume and revenue across categories like Mini, Prime, Auto, Bike, and E-Bike.
- **Cancellation Analysis**
  Highlights reasons for cancellations from both customers and drivers, helping identify major problem areas.
- **Ratings Analysis**
  Displays customer and driver ratings to assess service quality and overall satisfaction.

Interactive filters and slicers are included for dynamic exploration, allowing users to drill down into specific time periods, booking statuses, and vehicle categories.

- ## Key Insights

  - **Strong Demand with High Booking Volume**
  The platform recorded around **100K total bookings**, which clearly shows strong user demand and market presence. This indicates that the service is widely adopted and has consistent usage across the selected time period.

  - **Majority of Revenue Comes from Successful Rides**
  Out of all bookings, **63% were successful**, and these rides contributed the largest share of total revenue (~28M). This highlights that business growth is directly dependent on improving ride completion rates.

  - **High Cancellation Rate is a Major Concern**
  Approximately **27% of bookings were cancelled**, which is quite significant. This suggests operational inefficiencies such as driver unavailability, long waiting times, or pricing issues that need immediate attention.

  - **Customer vs Driver Cancellation Patterns**
  Cancellation analysis shows that both customers and drivers contribute to ride cancellations, but the reasons differ. Customers may cancel due to delays or pricing, while drivers may cancel due to distance or ride mismatch. This indicates a need for better matching algorithms and communication.

  - **Balanced Contribution Across Vehicle Types**
  Different vehicle categories like Mini, Prime, Auto, and Bike show relatively **balanced usage and revenue contribution**, meaning the platform is not dependent on a single segment and has diversified demand.

  - **Peak Demand Observed on Specific Days**
  Booking trends show noticeable spikes on certain days, indicating **peak demand periods (likely weekends or busy hours)**. This insight can help in better driver allocation and surge pricing strategies.

  - **Consistently High Ratings Reflect Good Service Quality**
  Both customer and driver ratings are around **4.2–4.3**, which reflects a generally positive user experience. Despite cancellations, the service quality remains strong for completed rides.

  - **Revenue Opportunity Hidden in Failed Rides**
  A large number of cancelled and incomplete rides represent **lost revenue opportunities**. Reducing cancellations even by a small percentage could significantly increase total revenue.

- ## Recommendations

  - **Reduce Cancellation Rate (Top Priority)**
  Focus on identifying root causes of cancellations (driver delays, pricing, availability) and implement solutions like better driver allocation and real-time updates to users.

  - **Improve Driver Allocation System**
  Optimize the ride-matching algorithm to assign nearby drivers more efficiently, reducing waiting time and increasing ride success rate.

  - **Introduce Incentives for Drivers**
  Provide performance-based incentives to drivers for completing more rides and avoiding cancellations, especially during peak hours.

  - **Enhance Customer Experience**
  Improve app communication by showing accurate wait times, fare transparency, and real-time ride tracking to reduce user frustration and cancellations.

  - **Leverage Peak Demand Insights**
  Use peak-time data to implement surge pricing and ensure higher driver availability during high-demand periods.

  - **Focus on Cancellation Reason Analysis**
  Regularly monitor and analyze cancellation reasons (both customer and driver side) to take targeted actions and reduce repeat issues.

  - **Convert Lost Opportunities into Revenue**
  Even a small reduction in cancelled rides can significantly boost revenue, so focusing on improving ride completion rate should be a key business strategy.

- ## Conclusion

  Overall, the data tells a clear story - the demand is strong, users are engaged, and the service quality is good. But the gap lies in converting bookings into successful rides. Every cancelled ride is not just a lost trip, but a missed opportunity. By addressing this gap, the platform can move from good performance to truly exceptional growth.

  However, the study also highlights a critical challenge in the form of a high cancellation rate, which directly impacts both revenue and user experience. A significant portion of potential business is being lost due to incomplete or cancelled rides.